TECHNICAL BLUEPRINT & CREATIVE PRODUCTION MANUAL

MCP Server Flow & Storyboard Production Bible

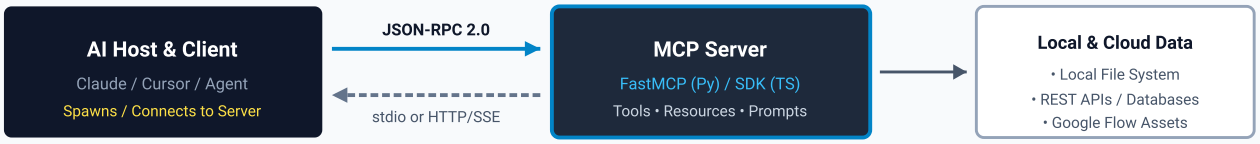
Part I: Step-by-Step Model Context Protocol Architecture • Part II: Google Flow Music Video Storyboard

1. CORE MCP ARCHITECTURE & PRIMITIVES

1. Resources (Read-Only)
URI-addressed data sources (files, database records, style guides) exposed to the AI for context, acting like `GET` endpoints.

2. Tools (Executable)
Typed functions the LLM actively invokes to compute values, call external APIs, or mutate state, acting like `POST` endpoints.

3. Prompts (Templates)
Pre-built, parameterized prompt workflows and slash commands that standardize recurring user tasks.



2. STEP-BY-STEP GENERIC MCP BUILD FLOW (STEPS 1–4)

STEP 1 **Define Scope & Transport Layer:** Map out your exact *Tools*, *Resources*, and *Prompts*. Select `stdio` (Standard Input/Output) for local CLI subprocesses or `SSE / Streamable HTTP` for remote cloud deployments.

STEP 2 **Scaffold Project & Install SDK:** Initialize an isolated environment using Python 3.10+ (`uv init my-mcp-server && uv add "mcp[cli]"`) or Node.js 18+ (`npm install @modelcontextprotocol/sdk zod`).

STEP 3 **Instantiate the Server:** Create `server.py` (or `index.ts`) and initialize the server object (`mcp = FastMCP("ServerName")`) to manage capability negotiation during the initial handshake.

STEP 4 **Register Capabilities with Strict Types:** Attach decorators (`@mcp.tool()`, `@mcp.resource()`, `@mcp.prompt()`). Provide explicit type hints and clear docstrings—the SDK compiles these directly into JSON Schema so the LLM knows when and how to invoke them.

3. REFERENCE IMPLEMENTATION & HOST REGISTRATION (STEPS 5–7)

```python
# Python Reference Implementation (server.py) using FastMCP
from mcp.server.fastmcp import FastMCP

mcp = FastMCP("StoryboardProductionServer")

# 1. Tool: Executable calculation or external API action
@mcp.tool()
def calculate_scene_frames(duration_seconds: float, fps: int = 24) -> int:
    """Calculate exact frame count for an animation shot given duration and FPS."""
    return int(duration_seconds * fps)

# 2. Resource: Read-only style & character consistency lock
@mcp.resource("config://studio/character-lock")
def get_character_lock() -> str:
    """Return the default character consistency anchor and hex palette."""
    return "Lead: 5yo Leo | Coat: #FFD700 | Boots: #D2143A | Style: 3D Pixar-style"

# 3. Prompt: Reusable verification template
@mcp.prompt()
def review_shot_prompt(shot_text: str) -> str:
    """Check a shot prompt against character consistency rules."""
    return f"Verify this shot includes the Character & Style Anchors:\n\n{shot_text}"

if __name__ == "__main__":
    mcp.run(transport="stdio")
```

STEP 5 **Bind Transport Entrypoint Safely:** Call `mcp.run(transport="stdio")`. *Critical Rule:* Never use `print()` to `stdout` in `stdio` mode, as it corrupts JSON-RPC frames. Log exclusively to `sys.stderr` or a file.

STEP 6 **Verify with the MCP Inspector:** Run `mcp dev server.py` (or `npx @modelcontextprotocol/inspector`) to test schema discovery (`tools/list`) and simulate payload execution (`tools/call`) in a local browser UI before connecting an LLM.

STEP 7 **Register in AI Host Config:** Add the absolute path of your project to your client's configuration file (e.g., `claude_desktop_config.json` or Cursor MCP settings):

```json
{
  "mcpServers": {
    "storyboard-server": {
      "command": "uv",
      "args": ["--directory", "/absolute/path/to/my-mcp-server", "run", "server.py"]
    }
  }
}
```

4. JSON-RPC 2.0 RUNTIME LIFECYCLE

Phase	Direction	JSON-RPC Method	Runtime Behavior
1. Handshake	Client → Server	`initialize`	Exchanges protocol versions and declares supported server capabilities.
2. Discovery	Client → Server	`tools/list`, `resources/list`	Fetches JSON schemas and docstrings to inject into the LLM context window.
3. Invocation	LLM → Client → Server	`tools/call`	LLM selects a tool and emits structured arguments; Client forwards request.
4. Execution	Server → Client → LLM	`result` / `error`	Server executes Python/TS code and returns output for final LLM synthesis.

PART II • VISUAL PRODUCTION BIBLE & ASSET LOCK

"Yellow Coat and Red Boots" — Storyboard Guide

Character Model Sheet, Environmental Lighting Scripts & Color Palette for Google Flow

1. CHARACTER MODEL SHEET & RECURRING MOTIFS

Lead Character: "Leo" (Age 5)

- **Proportions:** Stylized preschool build (1:3.5 head-to-body ratio), soft round face, rosy peach cheeks, button nose.
- **Hair & Eyes:** Chestnut-brown tousled hair with crown cowlick; large oval dark-brown eyes with bright specular highlights.
- **Outfit 1 (Indoor Cozy, 0:00–0:41):** Soft heather-grey cotton crewneck, navy jogger pants, cream fuzzy knit grip socks.
- **Outfit 2 (Outdoor Rain, 0:41–2:30):** Matte canary-yellow (■ #FFD700) hooded raincoat with 3 navy toggles, dark denim dungarees, high-gloss cherry-red (■ #D2143A) mid-calf rubber boots.

Signature Props & Visual Anchors

- **The Origami Boat:** Folded from heavy cherry-red (#D2143A) craft paper matching the boots; floats slightly tilted on puddle surfaces.
- **Clear Dome Umbrella:** Transparent canopy with cherry-red trim and curved canary-yellow J-handle (featured in Chorus 2).
- **Animal Cameo:** Small plump garden robin with an orange breast perched on a white wooden fence post, tilting its head at splashes.
- **Core Chorus Hook:** *"Rain, rain, fall today / We will splash and play / Splish and splash around / On the puddle ground"*

2. ENVIRONMENT & LIGHTING DIRECTION

Setting A — Indoor Window Nook (0:00–0:41): Low oak window bench framed by sheer white curtains. Dual-temperature split lighting pairs a warm 2800K bedside lamp glow inside with cool 6500K diffuse slate-blue rainy daylight outside. Diagonal rain streaks create soft garden bokeh.

Setting B — Rain-Washed Garden & Sidewalk (0:41–2:13): Suburban yard with high-specularity wet dark asphalt driveway, saturated emerald lawn, and white picket fence. Shadowless pearl-grey overcast sky (#C8D1D9) makes the canary-yellow coat and cherry-red boots pop vividly.

Setting C — Sunburst Resolution (2:13–2:30): Volumetric 3200K golden-hour sunrays pierce parting pewter clouds. Faint steam rises off drying asphalt, puddles turn to liquid gold, and a soft prismatic rainbow arcs across the upper-right sky.

3. MASTER COLOR SCRIPT & CAMERA TREATMENT

Song Phase	Dominant Palette	Key Accents	Camera & Lens Style
Intro & Verse 1 (0:00 – 0:41)	■ Slate Blue #4A607A ■ Warm Cream #FDF6E2	■ Amber Glow #FFB347 ■ Oak Wood #8B5A2B	Static framing, gentle push-ins, shallow depth of field.
Chorus 1, V2, Ch 2 (0:41 – 2:13)	■ Asphalt #2F3542 ■ Cloud Grey #A4B0BE	■ Canary #FFD700 ■ Cherry Red #D2143A	Low knee-height tracking, high shutter speed to freeze splashes.
Outro Resolution (2:13 – 2:30)	■ Sky Azure #70A1FF ■ Fresh Green #7BED9F	■ Sun Gold #FFA502 Prismatic Rainbow	Wide crane tilt upward toward sky, warm blooming highlights.

4. SUB-10-SECOND TIMESTAMP STORYBOARD & GOOGLE FLOW PROMPT MATRIX

Mandatory Google Flow Prompt Anchors (Prepend to every shot generation):
• **Character Anchor:** `5-year-old child Leo, messy tousled chestnut-brown hair with cowlick, large curious oval dark-brown eyes, round rosy peach cheeks`
• **Style Anchor:** `Pixar-style 3D stylized animation, soft warm lighting, tactile textures, rich color saturation, 8k resolution, cinematic composition`

Timestamp	Shot & Section	Camera & Visual Action	Google Flow Keyframe Prompt Specification
0:00–0:08	Shot 1 (Intro)	Eye-level close-up; rain on windowpane.	Cozy bedroom window, diagonal rain streaks on glass, blurred garden bokeh, warm 2800K amber desk lamp on oak sill vs cool 6500K rain light.
0:08–0:16	Shot 2 (Verse 1A)	Medium close-up; Leo watches clouds.	`[Character Anchor]` in heather-grey crewneck, hands resting on window glass, smiling up at soft grey rain clouds, split warm/cool lighting.
0:16–0:24	Shot 3 (Verse 1B)	Extreme macro; droplet on green leaf.	Glittering raindrop splashing onto broad emerald hosta leaf, concentric water ripples, blurred rainy garden background, high tactile detail.
0:24–0:32	Shot 4 (Verse 1C)	Low-angle exterior; roof & downspout.	Rain pattering rhythmically on dark grey roof shingles and gushing from white gutter spout against pearl-grey sky (#C8D1D9).
0:32–0:41	Shot 5 (Verse 1D)	Wide interior; wrapped in cozy blanket.	`[Character Anchor]` wrapped snugly in cream knit blanket on oak window seat, peaceful smile watching rain streaks outside.
0:41–0:50	Shot 6 (Chorus 1)	Dynamic low-angle; suiting up in gear.	`[Character Anchor]` zipping canary-yellow (#FFD700) raincoat with navy toggles and stepping into glossy cherry-red (#D2143A) boots.
0:50–0:58	Shot 7 (Chorus 1)	Full-body action track; first puddle leap.	`[Character Anchor]` in yellow coat (#FFD700), denim dungarees, and red boots (#D2143A) leaping mid-air into driveway puddle, big splash.
0:58–1:07	Shot 8 (Verse 2A)	Ground-level macro slow-mo stomp.	Cherry-red (#D2143A) rubber boots stomping puddle on dark wet asphalt, frozen arcs of sparkling droplets, yellow coat reflection.
1:07–1:16	Shot 9 (Verse 2B)	Medium shot; twirling under grey sky.	`[Character Anchor]` twirling gleefully with arms wide on wet sidewalk in yellow coat and red boots, fluffy storm clouds overhead.
1:16–1:24	Shot 10 (Verse 2C)	High-angle downward; curbside stream.	Miniature rapid river of clear rainwater flowing along dark asphalt street curb, carrying tiny floating green leaves.
1:24–1:32	Shot 11 (Verse 2D)	Knee-level profile; launching paper boat.	`[Character Anchor]` crouching by puddle in yellow coat, launching folded cherry-red origami paper boat; orange-breasted robin on fence post.
1:32–1:41	Shot 12 (Chorus 2)	Side tracking; high-stepping march.	`[Character Anchor]` marching energetically with high knees through shallow puddles, red boots splashing to beat, laughing expression.
1:41–1:48	Shot 13 (Chorus 2)	Low-angle upward; clear umbrella spin.	`[Character Anchor]` spinning clear dome umbrella with red trim and yellow J-handle, raindrops bouncing off canopy, mist on rosy cheeks.
1:48–1:57	Shot 14 (Bridge)	Ground-level montage; dancing drops.	Sparkling raindrops bouncing like beads on mirror-like puddle; curious garden robin chirping on white picket fence in background.